HUMAN-AI INTERACTION COURSE | PROJECT MILESTONE

Academic Writing Coach

An Interactive AI Assistant for University Students Focusing on Transparency, Grounded Feedback, and Uncertainty Handling.

1. Project Overview & Context

The Student Challenge

University students frequently struggle with academic phrasing, formal tone, structural coherence, and proper citation formats (APA, MLA, IEEE) when drafting term papers and research reports.

Generic AI tools often provide ungrounded rewrites or plausible-sounding fabrications (AI hallucinations), undermining academic integrity.

Our Solution & Transparency

The **Academic Writing Coach** acts as an interactive pedagogical consultant. Students submit draft excerpts and receive constructive feedback.

Every critique is anchored to official style guides. When rules are ambiguous, the system explicitly discloses uncertainty.

2. Fictional User Personas

Persona 1

Youssef Ibrahim

Below-Typical Tech Level & Accessibility Needs

First-year student unfamiliar with AI tools. Needs high contrast, plain-language explanations, and clear citation badges without dense stylistic jargon.

Focus: Accessibility & Simplicity

Persona 2

Mariam Adel

Typical Technical Level

Second-year student who understands basic essay structure but struggles with proper in-text citation formatting and formal academic tone.

Focus: Rule Verification & Tone

Persona 3

Ziad Khaled

Advanced Technical Level

Third-year student experienced in research synthesis. Seeks direct style manual citations and advanced structural diagnostics instantly.

Focus: Granular Diagnostics

3. Bias Check & Data Source Design

Bias-Check Highlights

- **Cultural Neutrality:** Technical proficiency and writing capability are decoupled from cultural backgrounds.
- **Accessibility-First:** UI design avoids color-only indicators, supporting cognitive load sensitivity.
- **Uncertainty Handling:** Prevents the system from enforcing subjective stylistic norms as absolute rules.

Synthetic Data Source

Utilizes a curated JSON dataset (`academic_guidelines_dataset.json`) targeting 40 structured cases covering grammar, citation rules, and structural arguments.

Contains zero real student personal data or private drafts, ensuring complete privacy compliance.

Milestone 1: Two AI Concept Drafts Side-by-Side

Draft A (ChatGPT)

Conversational Mentoring Flow

A question-first, guided-diagnosis approach where students submit text and receive incremental hints and conversational explanations before seeing fixes.

Strength: Intuitive & flexible. **Weakness:** Open-ended queries can overwhelm beginners.

Draft B (Gemini)

Rule-Based Citation & Style Board

A structured verification board where students select style manual criteria (APA/IEEE) to systematically validate references and structural guidelines.

Strength: Highly rigorous. **Weakness:** Can feel rigid for creative drafting.

Milestone 1: Direction & Unique Scope Approval

Chosen Direction & Rationale

We chose a **hybrid architecture** combining Conversational Mentoring (Draft A) with Rule-Based Citation Verification (Draft B). This grants students full control over their editing path while enforcing strict style manual grounding and transparency.

Unique Idea & Scope Approval

Target User Group: University students seeking academic writing support.
Unique Value: Unlike general helpdesks or coding mentors, our project specifically targets academic prose, stylistic critique, and style-guide grounding with explicit uncertainty handling.

4. End-to-End System Workflow

1. **Input & Start**
   Student submits draft paragraph and writing query via the responsive web interface.

2. **Grounding & Backend Verification**
   Python FastAPI backend validates the query against `academic_guidelines_dataset.json`.

3. **Transparent Feedback Delivery**
   System returns constructive critique paired with official style guide badges or uncertainty notes.

4. **Action & Feedback**
   Student applies independent revisions and submits ratings to refine system performance.

5. User Journey Stages

1. Start

Anxiety over looming essay deadlines and citation rules.

2. Ask

Submitting draft excerpts with plain-language queries.

3. Understand

Reviewing critique paired with style manual references.

4. Act / Fix

Following official style links to revise drafts independently.

5. Finish

Submitting revised essays with total confidence and accuracy.

6. Measurable Objectives (Target Metrics)

O1 – Rule & Citation Accuracy	≥ 90% Accuracy
O2 – Task Success Rate	≥ 80% Success
O3 – Transparency Comprehension	≥ 90% Comprehension
O4 – Performance Latency	< 2.0 Seconds
O5 – Uncertainty Handling Rate	≥ 95% Fallback Rate

7. HAI Principle Ladder — Transparency

Level 1 — Inform

Discloses AI assistant scope and capabilities upon opening the chat interface.

Level 2 — Explain (Rule Attribution)

Attaches direct style manual links and references to every structural critique.

Level 3 — Allow Control (Uncertainty Disclosure)

Displays explicit uncertainty warnings when stylistic rules are ambiguous or open to instructor discretion.

Level 4 — Enable Feedback

Collects user ratings on suggestion clarity and helpfulness to continuously refine the mentor.

8. Technical Architecture & Evaluation

Technical Architecture

- **Frontend:** HTML5, CSS3, and JavaScript with accessible card layouts and screen-reader support.
- **Backend:** Python FastAPI handling request routing and rule validation.
- **Grounding:** Hosted LLM API integrated with `academic_guidelines_dataset.json` to eliminate hallucinations.

Evaluation Plan

- **Technical Testing:** Automated script verification comparing feedback against official style manuals.
- **Usability Testing:** Peer testing with at least 5 student participants performing essay revisions.
- **SUS Surveys:** System Usability Scale questionnaires administered to evaluate interface usability.

Thank You!

Questions & Discussion